Dr. A.P.J. Abdul Kalam: The Missile Man

Introduction

Dr. Avul Pakir Jainulabdeen Abdul Kalam, widely known as A. P. J. Abdul Kalam, was an Indian aerospace scientist, teacher, author and the 11th President of India. He was affectionately called the "Missile Man of India" and the "People's President". Born into a modest family, he rose to become one of the most admired Indians of all time through hard work, curiosity and a deep love for teaching. His life remains a source of inspiration to millions of students.

Early Life: Where Was Kalam Born?

Kalam was born on 15 October 1931 in Rameswaram, on Pamban Island in the Ramanathapuram district of the Madras Presidency, now Tamil Nadu. His father, Jainulabdeen, owned boats and served as an imam of a local mosque, and his mother Ashiamma was a homemaker. The family was not wealthy, and the young Kalam sold newspapers to help with the family income. He studied at the Rameswaram Elementary School and Schwartz Higher Secondary School in Ramanathapuram, then went on to study physics at St. Joseph's College in Tiruchirappalli, graduating in 1954. He then studied aerospace engineering at the Madras Institute of Technology (MIT) in Chromepet, Chennai, and graduated in 1960. His teacher Sivasubramania Iyer and his cousin Samsuddin influenced him, and he loved flying and birds.

Career at DRDO and ISRO

Kalam joined the Defence Research and Development Organisation (DRDO) in 1958 and worked on a small hovercraft, and then moved to the Indian Space Research Organisation (ISRO) in 1969. At ISRO, he was the project director of India's first indigenous satellite launch vehicle, the SLV-3, which successfully placed the Rohini satellite in orbit on 18 July 1980, making India a member of the exclusive club of space-faring nations. He worked with Dr. Vikram Sarabhai and later with Satish Dhawan, whom he admired as a mentor.

Why Is He Called the Missile Man?

Kalam is called the Missile Man because he led the development of India's ballistic missile and launch vehicle technology. In 1982 he returned to DRDO as the director of the Defence Research and Development Laboratory, and he was given the lead of the Integrated Guided Missile Development Programme (IGMDP), started in 1983. Under this programme India developed a family of missiles: Prithvi (surface-to-surface), Agni (intermediate-range ballistic missile), Trishul and Akash (surface-to-air) and Nag (anti-tank). The first Agni missile was successfully tested in May 1989, a milestone for India's defence. Kalam combined space-rocket and missile knowledge and built teams of young scientists. He served as Scientific Adviser to the Defence Minister and Secretary of DRDO from 1992 to 1999, and also as Principal Scientific Adviser to the Government of India from 1999 to 2001, and he played a key role in the Pokhran-II nuclear tests in May 1998, along with Dr. R. Chidambaram. His work on medical devices included a low-cost coronary stent and a lightweight carbon-fibre calliper for polio patients, developed with a cardiologist, Dr. Soma Raju.

When Was He President of India?

Dr. Kalam served as the 11th President of India from 25 July 2002 to 25 July 2007, a five-year term. He was nominated by the National Democratic Alliance government and was supported by the opposition as well, winning the election in July 2002 by a big margin over Captain Lakshmi Sahgal. He was the first scientist and the first bachelor to occupy the Rashtrapati Bhavan. He was known as the "People's President" because he was simple, open and spent time meeting people and young students; he held interactions with lakhs of students during his term. He promoted the idea of PURA (Providing Urban Amenities in Rural Areas) to bring urban facilities to villages.

Honours

He received India's highest civilian award, the Bharat Ratna, in 1997, as well as the Padma Bhushan in 1981 and the Padma Vibhushan in 1990, along with more than 40 honorary doctorates from universities around the world. In 2010 the United Nations declared his birthday, 15 October, as World Students' Day.

Books and Vision

Kalam authored several books, including his autobiography "Wings of Fire" (1999), written with Arun Tiwari, as well as "India 2020: A Vision for the New Millennium" (1998), written with Y. S. Rajan, "Ignited Minds", "My Journey" and "Turning Points". He urged India to become a developed nation by 2020 through his vision of Vision 2020 and encouraged young people to dream, saying that dreams should be turned into thoughts and thoughts into actions. He spent many years as a professor and visiting lecturer at institutions such as Anna University in Chennai, IIM Ahmedabad and IIM Indore, and he launched the "What Can I Give" movement for youth.

Death and Legacy

Dr. Kalam passed away on 27 July 2015, at the age of 83, after suffering a cardiac arrest while delivering a lecture at the Indian Institute of Management in Shillong. He was buried at Pei Karumbu in Rameswaram with full state honours, and his memorial, the Dr. A. P. J. Abdul Kalam National Memorial, was opened there in 2017. Many institutions, including the Dr. A. P. J. Abdul Kalam Island at Wheeler Island in Odisha, where missiles are tested, were named after him, and Tamil Nadu celebrates his birthday as Youth Renaissance Day.

Quick Facts

- Kalam was born on 15 October 1931 in Rameswaram, Tamil Nadu.
- He is called the Missile Man for leading India's Integrated Guided Missile Development Programme, with the Agni and Prithvi missiles.
- He was India's 11th President from 25 July 2002 to 25 July 2007.
- He led the SLV-3 project that launched the Rohini satellite on 18 July 1980.
- Wings of Fire is his autobiography; he was awarded the Bharat Ratna in 1997.